

A hybrid grey approach for battery remaining useful life prediction considering capacity regeneration

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ARTICLE INFO

Keywords:

Remaining useful life
Lithium-ion battery
Grey forecasting model
Ensemble Kalman Filter

ABSTRACT

Remaining useful life (RUL) prediction is the core of prognostic and health management. Lithium-ion batteries are a kind of consumable resource, whose RUL has a revelatory effect on its safe use and management. However, it has been found that battery degradation is not a completely degrading trend, which is ignored by traditional prediction methods. This paper proposes a hybrid grey approach to predict the RUL under multiple states. To predict the detailed degradation as real as possible, the capacity regeneration phenomenon is found to be a crucial part. A hybrid grey forecasting model is established to forecast the normal degradation and capacity regeneration process in detail. Ensemble Kalman Filter (EnKF) algorithm is applied to weaken the errors of grey forecasting models for long-term prediction. Based on single models from grey modeling and filtering, the ablation study shows the influence of different components in the hybrid approach. Then, compared with other data-driven models, the hybrid grey-EnKF demonstrates a quite satisfactory prediction of RUL for a set of batteries. Results show the grey-EnKF predictions are highly accurate with mean absolute percentage error smaller than 1%.

1. Introduction

Lithium-ion batteries, as common tools in manufacturing systems, are a consumable resource. An accurate understanding of the battery RUL enables better battery health management for its safety, reliability and economics. With increasing attention paid to battery health, more and more prediction methods are developed (Alsuwian et al., 2024; Dai et al., 2024). Here, we divide these methods into three types. Model-based approaches usually establish a mathematical model such as an exponential model based on the electrochemical reaction. In the prevalence of big data, data-driven approach fully seeks information from the data so that it does not analyze the system mechanism. Compared with model-based approaches, it is more feasible and practical. Hybrid approaches that integrate model-based and data-driven approaches even over two model combinations have also been proposed for higher accuracy.

In mathematical statistics, stochastic process, a kind of model-based approach, is one of the important methods in RUL prediction (Wang et al., 2019). Wiener process considers the randomness and individual difference (Lin et al., 2019). A two-phase model is proposed along with an estimation method for the change point and model parameters (Liu et al., 2022) and it also combines with Markov model (Mu et al., 2021). Fractional Brownian Motion model is introduced to RUL

mechanism (Wang et al., 2020). Subsequently, a data-fusion method is developed to establish a non-parametric degradation model (Li et al., 2022). On the basis of single-stage prediction, multi-stage prediction is proposed when joint classification is conducted at first. Under the regression algorithm, RUL predictions are performed at stage level (Wu et al., 2021). In an uncertainty system, deep learning and non-stationary Gaussian process regression are integrated due to their superiority (Ansari et al., 2024; Li et al., 2019; Xu et al., 2022a). For individual units, a statistical prognosis based on noisy condition monitoring is applied (Son et al., 2016; Zhang et al., 2024b).

During battery degradation process, it is difficult to monitor the current, voltage, capacity and other relative indexes so it is hard to find degradation mechanisms (İslim & Çatay, 2024; Thomas et al., 2019). Machine learning is especially popular for researchers (Cai & Lin, 2024; Ferreira & Gonçalves, 2022; Ghosh et al., 2023; Guo et al., 2024). For example, a convolutional network is an important tool with Bayesian optimization. In the adversarial training framework, the temporal convolutional network is a generator (Chen et al., 2022). Convolutional autoencoder networks are also combined with state degradation (Huang et al., 2021; Xu et al., 2022b). To improve the accuracy for long-time prediction, a deep belief networks method

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is proposed (Han et al., 2022; Tang et al., 2022). For hybrid approaches, the Wiener process and LSTM network are integrated after data augmentation processing (Wang et al., 2022). Empirical mode decomposition (EMD) is an adaptive method for analyzing nonlinear and non-steady signals. In recent years, many improved EMDs have been used to predict the battery RUL. To eliminate the uncertainty caused by capacity regeneration, EDM is applied and the capacity time series is clearly decomposed into multiple intrinsic mode functions and residual (Sun et al., 2024). The intrinsic pattern function represents the uncertain changes brought about by capacity regeneration, while the residual function represents the overall trend. High-frequency noise signals are filtered through the EMD algorithm and the Pearson correlation coefficient (Yao et al., 2023). Adopt a complete EEMD to decompose adaptive noise to achieve complete adaptive decomposition of original data, which prevents the effective information about the capacity regeneration from being eliminated (Tang & Yuan, 2022). At the same time, error compensation is also used in RUL (Guo & He, 2022). Filtering is a common tool for RUL prediction (Takyyi-Aninakwa et al., 2024). Kalman filter (KF) has been developed for the state estimation of linear dynamic systems. Extended Kalman filter (EKF) and unscented Kalman filter (UKF) techniques are used to state of charge estimation for battery (Priya et al., 2023). Combined with an improved equivalent circuit model, a state of charge evaluation scheme is proffered (Zhang et al., 2024). Based on improved particle swarm optimization-adaptive square root cubature KF, a novel method is proposed to reduce the trial and error time of artificial selection. In the theory of particle filter (PF) algorithm, its extended form is developed (Xie et al., 2020). Unscented particle filter, one of its extensions, is introduced to the battery RUL prediction (Miao et al., 2013). With the help of PF, traditional time series models and statistical analysis are used to track the battery degradation during its charging and discharging (Ahwiadi & Wang, 2022). To realize information interaction, support vector regression is conducted for multi-step-ahead estimation with the continuous iteration of filtering algorithm (Li et al., 2021). Besides, more and more interacting multiple models are developed (Chen et al., 2021b; He et al., 2022).

Due to the difference between the kinds of batteries, a mass of data collected by the monitor does not have a big influence on the RUL prediction, which also has many costs. Grey forecasting model, like the other data-driven methods, occupies a place in forecasting methods. The biggest advantage of grey models is that their modeling mechanism is based on small data and they mine information hidden limited clutter of data. Grey models not only describe capacity degradation but also revise their prediction errors (Chen et al., 2015). For different health indexes in battery, the primary task is to identify the most effective indicators that reflect degradation (Zhou et al., 2017). Then GM(1,1) and grey Verhulst model are applied with particle swarm optimization and PF (Xian et al., 2014).

However, these methods only focus on the accuracy of the RUL prediction for many alternative methods in the application and pay less attention to the detailed degradation process. Especially, there is a regeneration phenomenon during the whole degradation (Ma et al., 2021; Wang, 2019). Capacity regeneration often occurs when the battery is standing. Since the use is not continuous, there is an interval between each charge and discharge, and the emergence of capacity regeneration has a great impact on the use life of the entire battery. A single forecasting model can predict the degradation trend but not reflect the details of the degradation process. More and more grey prediction models with different structures appear, while there are few studies on mining models applicable to sequences according to their modeling mechanisms. Although the accuracy of the model improves with the increase and adjustment of nonlinear parameters, the more complex model structure lacks certain explanations. And the performance of the grey prediction of certainty will be greatly reduced with the increase of the predicted step length. In this paper, grey forecasting models are selected as the main prediction model combined with EnKF

algorithm. Therefore, the hybrid grey-EnKF approach is proposed. The main contributions of this paper are as follows:

(1) Considering the details of the real degradation process, the grey forecasting model is the first time used to deal with battery RUL prediction with capacity regeneration phenomena. Different from frameworks of data-driven methods, it is satisfying for grey forecasting models based on the simple modeling mechanism that few data can provide abundant information.

(2) According to degradation characteristics in different stages, hybrid grey forecasting models are established to forecast the normal degradation and capacity regeneration process, which maximizes the powerful advantages of grey modeling in mining sequence evolution. Combined with EnKF, long-term forecast error is weakened by continuous updates.

(3) Based on battery public datasets, prediction performances of the hybrid grey-EnKF approach are demonstrated and analyzed. Ablation study shows the influence of components of grey-EnKF. Moreover, comparative analysis with single models and popular machine learning is conducted.

The rest of paper is arranged as follows. Section 2 analyzes the battery capacity degradation behavior. Section 3 presents the modeling mechanism, construction and modeling process of the hybrid approach. In Section 4, the application of RUL prediction based on a set of lithium-ion batteries is carried out. A comparison with single models, algorithms and existing prediction methods is shown. Finally, conclusions are summarized in Section 5.

2. Battery capacity degradation analysis

Under normal conditions, the battery capacity has degraded by repeated cycles of charge and discharge over time. However, according to data on batteries provided by NASA in Fig. 1, it is not a complete degrading process. In part of cycles, the capacity appears to be more than that at the previous cycle. This phenomenon is due to the weak and complex electrochemical reactions that occur inside the battery when it does not work, which makes the capacity recover in a short time. The existence of this phenomenon is relatively common, and this special situation cannot be directly ignored in quantitative RUL prediction.

To quantify the degradation of the battery specifically and accurately, define the state of health (SOH) of batteries as

$$SOH = \frac{C_k}{C_0} \quad (1)$$

where C_k is the battery capacity at the k th charge-discharge cycle (unit: Ah). C_0 indicates the rated capacity of the battery (Ah). The difference in SOH is an accurate indicator of capacity recovery. When $DH_k = SOH_k - SOH_{k-1}$ is greater than zero, it indicates that the capacity of the previous cycle has increased. However, in order to avoid the influence of measurement error in the monitoring system on the judgment of capacity regeneration, the threshold is set as 0.2%, that is, when $SOH_k - SOH_{k-1} > 0.2\%$, it is seen as the capacity regeneration. The regeneration interval is defined as the interval in which the capacity degrades at a faster rate after the capacity regeneration until it is less than the capacity of the $k-1$ cycle before the regeneration.

After the above definition, the complex battery capacity degradation is divided into normal steady decline and capacity regeneration. Take B5 as an example, its multi-state degradation process in Fig. 2. Further, we decompose the capacity degradation process. The normal degradation process presents a steadily decreasing trend. During the whole life cycle, battery during the normal degradation process degrades slowly at first, then accelerates, and then degrades slowly when the degradation reaches 70% of the rated capacity. This is consistent with the degradation mechanism of batteries. Meanwhile, the extracted capacity regeneration phenomenon is shown in the subfigure. Compared with the normal stable degradation, the degradation rate of battery capacity becomes faster after capacity regeneration occurs, especially in the

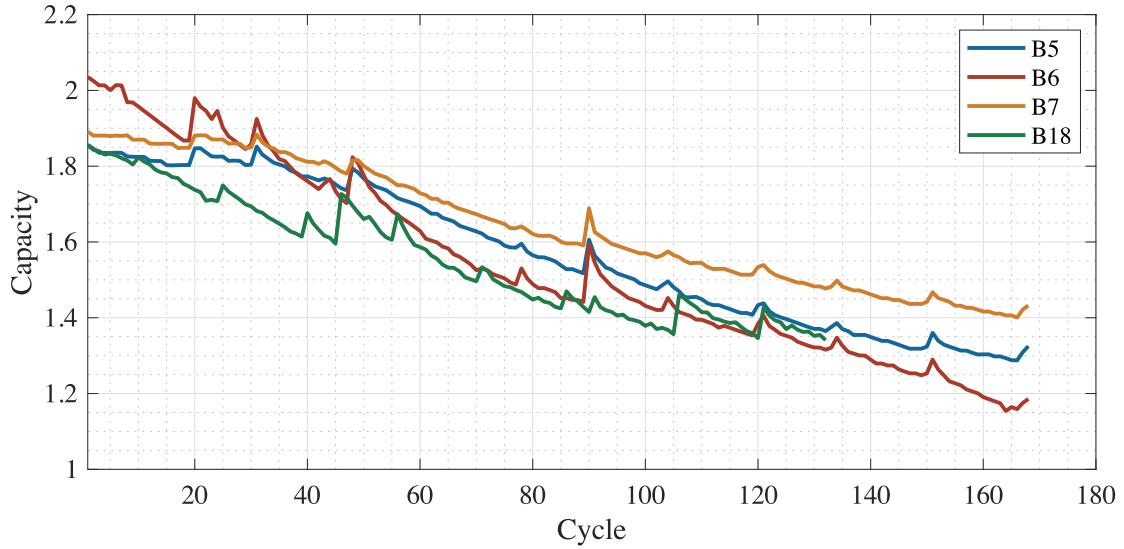


Fig. 1. Battery capacity provided by NASA.

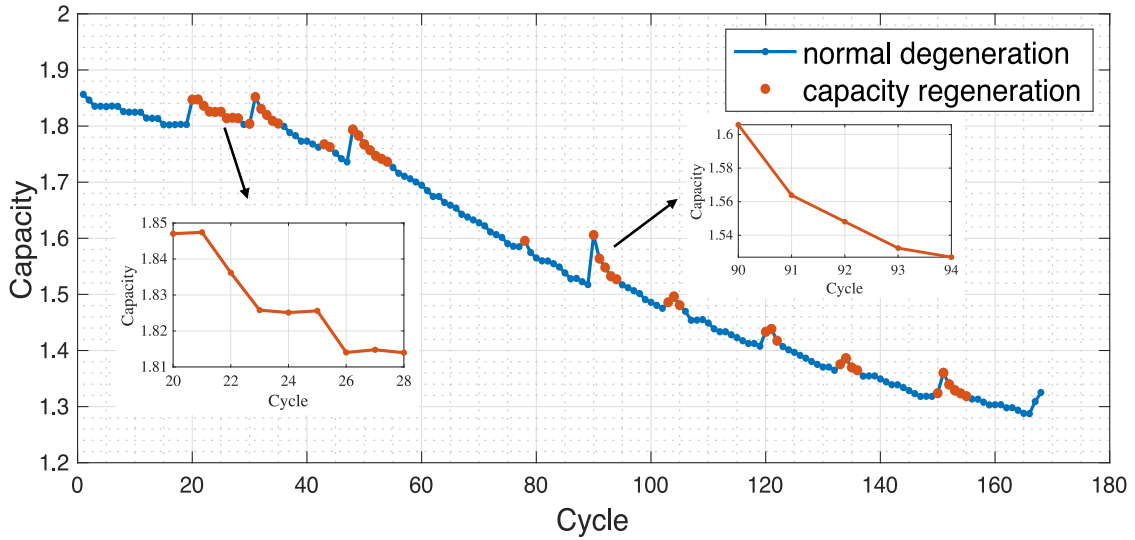


Fig. 2. Degradation analysis.

regeneration interval of the 20th, 31st, 43rd and 90th cycles, and the degradation capacity roughly declines non-homogeneously.

Through the analysis of battery capacity degradation behavior, it is found that the battery capacity is not an ideal continuous decay process, which is ignored by the traditional prediction methods, thus affecting the prediction accuracy. To a certain extent, the existence of capacity regeneration extends the service life of the whole battery. At the same time, the degradation processes under normal degradation and capacity regeneration show different degradation characteristics, which should be paid more attention to RUL prediction.

3. Methodology

In this section, the novel hybrid grey-EnKF approach is proposed. 'Grey' indicates that the grey forecasting model is built according to the trend of stable capacity degradation and regenerative features. 'EnKF' refers to the algorithm to weaken the errors of grey forecasting models for long-term prediction. Its modeling mechanism is detailed in Section 3.1. The construction of hybrid grey-EnKF is shown in Section 3.2, and detailed modeling steps are described in Section 3.3.

3.1. Modeling mechanism of the hybrid grey-EnKF approach

Due to the complex electrochemical reactions in the battery during the charge-discharge cycle, it is difficult to accurately describe the lithium-ion battery with the model based on physical mechanism. Data-driven prediction methods show advantages in this respect, avoiding the analysis of complex electrochemical reactions. As a data-driven grey forecasting model, it can be modeled based on small sample data, compared with the large number of data samples required by popular machine learning methods. Due to the randomness of the battery capacity regeneration phenomenon, the data that can be collected is more scarce. The grey forecasting model becomes a more ideal choice among many machine learning methods. Therefore, based on the analysis of battery capacity degradation behavior, a hybrid grey prediction model is established.

Firstly, the battery degradation showed a steady and nonlinear downward trend in the normal degradation stage. The normal degradation process degrades slowly at first, then accelerates, and then degrades slowly when it reaches 70% of the rated life. When the capacity is lower than 70% C_0 , the capacity gradually becomes stable. The overall trend is exponential. Under this condition, a grey forecasting

model is established for the feature sequence with a quasi-exponential growth.

Secondly, after the occurrence of the regeneration phenomenon, the degradation speed is accelerated with an exponential downward trend. Meanwhile, the occurrence of the regeneration phenomenon is relatively random, and the number of capable samples is small, which only accounts for less than 20% of the data in the whole degradation. For the capacity prediction under this state, a non-homogeneous discrete grey forecasting mode is adopted for prediction. It has higher flexibility than that for quasi-exponential sequences. The difference equation is directly established, which avoids the error caused by the jump in the internal parameters of the system.

Thirdly, it is idealized that there is no uncertainty in all systems in practical applications, so KF is widely used as an optimal recursive algorithm. EnKF is developed based on the KF to broaden the application scope. Using the idea of the set, the probability density distribution of Monte Carlo sample approximate state variables is run by the nonlinear numerical model, so that EnKF is no longer restricted by the noise that satisfies the Gaussian distribution and the linear model. Considering the uncertainty of the phenomenon of capacity regeneration and the inherent error of measurement, EnKF further compensates for the negative impact brought by long-term prediction of the grey forecasting model.

3.2. Construction of the hybrid grey-EnKF approach

Definition 1. Let $X^{(0)}$ be battery raw capacity data, $X_1^{(0)} = \{x_1^{(0)}(1), x_1^{(0)}(2), \dots, x_1^{(0)}(n_1)\}$ be the normal degradation process, and it satisfies Eq. (2). And $X_2^{(0)} = \{x_2^{(0)}(1), x_2^{(0)}(2), \dots, x_2^{(0)}(n_2)\}$ be the regeneration process.

$$\frac{x_1^{(0)}(i) - x_1^{(0)}(i-1)}{C_0} < 0.2 \quad (2)$$

Definition 2. Assume that $X^{(1)}, X_i^{(1)}$ are accumulation generated sequence of battery capacity data, $X_i^{(1)} = \{x_i^{(1)}(1), x_i^{(1)}(2), \dots, x_i^{(1)}(n_i)\}$, where $x_i^{(1)}(k) = \sum_{j=1}^k x_i^{(0)}(j)$. Let Z_i be the adjacent mean generation sequence, where $z_i(k) = \frac{1}{2}(x_i^{(1)}(k) + x_i^{(1)}(k-1))$.

Definition 3. Assume that $X_i^{(1)}, X_i^{(0)}$ are defined in Definitions 1 and 2,

$$\begin{cases} \frac{dx_i^{(1)}(t)}{dt} = f_1(x_i^{(1)}(t), u_i) \\ x_i^{(1)}(k) = f_2(x_i^{(1)}(k-1), u_i) \end{cases} \quad (3)$$

is called the hybrid grey forecasting model. Further, the basic form of discretization can be expressed as Eq. (4).

$$x^{(1)}(k) = ax^{(1)}(k) + kb + \beta_1 x^{(1)}(k-1) + \beta_2 k + \beta_3 \quad (4)$$

where $z_1^{(1)}(k)$ is the accumulation generated sequence of $z_1(k)$.

Theorem 1. Assume that Y_i, X_i, Z_i are defined in Definitions 1 and 2, the least squares estimation of the parameters $\hat{u} = (\hat{a}, \hat{b}, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3)^T$ satisfies

$$\hat{u} = (B^T B)^{-1} B^T D, \text{ where} \quad B = \begin{bmatrix} z_1(2) & 1 & 0 \\ z_1(3) & 1 & 0 \\ \vdots & \vdots & 0 \\ z_1(n) & 1 & 0 \\ x_2^{(1)}(1) & 1 & 1 \\ x_2^{(1)}(2) & 2 & 1 \\ \vdots & \vdots & \vdots \\ x_2^{(1)}(n-1) & n-1 & 1 \end{bmatrix}, D = \begin{bmatrix} x_1^{(0)}(2) \\ x_1^{(0)}(3) \\ \vdots \\ x_1^{(0)}(n) \\ x_2^{(1)}(2) \\ x_2^{(1)}(3) \\ \vdots \\ x_2^{(1)}(n) \end{bmatrix}, u = \begin{bmatrix} a \\ b \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \quad (5)$$

Theorem 2. Let Y_i, X_i, u_i be shown in Theorem 1, then

(1) The time-response formula of the hybrid grey model is

$$\begin{aligned} x^{(1)}(k) = & \left(x_1^{(0)}(1) - \frac{b}{a} \right) e^{-a(k-1)} + \frac{b}{a} + \beta_1^{k-1} x_2^{(0)}(1) + \beta_2 \sum_{j=1}^k j \beta_1^{k-j} \\ & + \frac{1 - \beta_1^{k-1}}{1 - \beta_1} \beta_3 \end{aligned} \quad (6)$$

(2) The prediction of battery capacity is

$$\hat{x}^{(0)}(k) = \hat{x}^{(1)}(k) - \hat{x}^{(1)}(k-1) \quad (7)$$

The known linear system model equation is as follows:

$$\begin{aligned} x_k^{(0)} &= A_{k-1} x_{k-1}^{(0)} + w_{k-1} \\ z_k &= B_k x_k^{(0)} + v_k \end{aligned} \quad (8)$$

where the $x_k^{(0)} \in R^n$ and $z_k \in R^n$ respectively are k th moment of state vector and measurement vector, $A_{k-1} \in R^{n \times n}$ and $B_k \in R^{n \times n}$ are the driving matrix and the measurement matrix, respectively. $w_{k-1} \in R^n$ and $v_k \in R^n$ respectively are process noise and measurement noise, the two mutually independent and the mean is 0, the respective covariance is Q_{k-1} and R_k , respectively.

Updating a priori estimation with new information $\hat{x}_{k|k-1}^{(0)}$. The posterior estimation is

$$\hat{x}_k^{(0)} = \hat{x}_{k|k-1}^{(0)} + G_k i_k = \hat{x}_{k|k-1}^{(0)} + G_k [z_k - B_k \hat{x}_{k|k-1}^{(0)}] \quad (9)$$

where, the G_k is the Kalman filter gain. Then the posterior estimation error of the Kalman filter is $e_k = x_k^{(0)} - \hat{x}_k^{(0)}$, further the error covariance matrix is $P_k = E[e_k e_k^T]$.

By a priori estimate error is $e_{k|k-1} = x_k^{(0)} - \hat{x}_{k|k-1}^{(0)}$, in the same way, a priori estimate error covariance matrix is $P_{k|k-1} = E[e_{k|k-1} e_{k|k-1}^T]$, then

$$\begin{aligned} e_k &= x_k^{(0)} - \hat{x}_k^{(0)} = x_k^{(0)} - \hat{x}_{k|k-1}^{(0)} - G_k (z_k - B_k \hat{x}_{k|k-1}^{(0)}) \\ &= x_k^{(0)} - \hat{x}_{k|k-1}^{(0)} - G_k (B_k x_k^{(0)} + v_k - B_k \hat{x}_{k|k-1}^{(0)}) \\ &= (I - G_k B_k) e_{k|k-1} - G_k v_k \end{aligned} \quad (10)$$

where I is the identity matrix. When setting $k=0$, a prior estimation is $\bar{x}$, then $P_k = E[e_k e_k^T] = 0$. Since the average of measurement noise is 0, also the mean of e_k also is zero, the covariance matrix $P_k = E[e_k e_k^T]$ can converse

$$P_k = E[e_k e_k^T] = (I - G_k B_k) P_{k|k-1} (I - G_k B_k)^T + G_k R_k G_k^T P_k. \quad (11)$$

The gain of Kalman filter with the minimum error can be obtained as:

$$G_k = P_{k|k-1} B_k^T (B_k P_{k|k-1} B_k^T + R_k)^{-1} \quad (12)$$

then we can obtain

$$P_k = P_{k|k-1} - P_{k|k-1} B_k^T (B_k P_{k|k-1} B_k^T + R_k)^{-1} B_k P_{k|k-1} = (I - G_k B_k) P_{k|k-1}. \quad (13)$$

Due to $E[A_{k-1}e_{k-1}w_{k-1}^T] = 0$, $E[w_{k-1}w_{k-1}^T] = 0$, then

$$P_{k|k-1} = E[A_{k-1}e_{k-1} + w_{k-1}^T] E[A_{k-1}e_{k-1} + w_{k-1}^T]^T$$

$$= A_{k-1}P_{k-1}A_{k-1}^T + Q_{k-1} \quad (14)$$

Updating hybrid grey model with EnKF. Under the EnKF framework, it is assumed that the input of the numerical model meets certain prior distributions (such as Gaussian distribution, exponential distribution, etc.), and then the Monte Carlo sampling method is used to obtain a sufficient number of input sample sets, and each sample is inserted into the known model to obtain the model output through the numerical operation.

Considering function

$$d_i = f(x_i^{(0)}) \quad (15)$$

where, $x_i^{(0)}$ represents the i th input sample, and N represents the number of samples.

The mean value and covariance of output can be obtained through the state equation in the hybrid grey forecasting model and it satisfies Eq. (16), where $\mu_m = \frac{1}{N} \sum_{i=1}^N x_i^{(0)}$.

$$\mu_d = \frac{1}{N} \sum_{i=1}^N f(x_i^{(0)})$$

$$C_{dd} = \frac{1}{N-1} \sum_{i=1}^N (f(x_i^{(0)}) - \mu_d)(f(x_i^{(0)}) - \mu_d)^T \quad (16)$$

$$C_{dm} = \frac{1}{N-1} \sum_{i=1}^N (f(x_i^{(0)}) - \mu_d)(m_i - \mu_m)^T$$

Each input sample is updated with the observation data

$$x_i^{(0)u} = x_i^{(0)} + K(f(x_i^{(0)})^* - f(x_i^{(0)})) \quad (17)$$

where $x_i^{(0)u}$ represents the i th input sample obtained by updating, d_i^* represents the i th observation data. K represents the Kalman gain matrix and it satisfies Eq. (18), where $C_{md} = C_{dm}^T$, R represents the covariance matrix of the observation error.

$$K = C_{md}(C_{dd} + R)^{-1} \quad (18)$$

3.3. Modeling steps of the hybrid grey-EnKF approach

Specific modeling steps are as follows and the framework is shown in Fig. 3.

Step 1: Obtain the battery capacity and data pre-processing.

Step 2: Judge whether capacity regeneration occurs by setting a threshold. According to the given threshold, the normal degradation interval and capacity regeneration interval of the battery are found and extracted from the original data.

Step 3: Group the modeling sequences, one for training and the other for testing.

Step 4: Compute the accumulation generated sequence of battery capacity and the adjacent mean sequence.

Step 5: Construct the hybrid grey forecasting model by Eq. (4).

Step 6: Parameters estimation using the least squares method according to Theorem 1.

Step 7: Solve hybrid grey forecasting model according to the estimated parameters in Step 6.

Step 8: Obtain the predicted values according to the inverse operator in Eq. (7).

Step 9: Update predicted value by EnKF. The time response of grey forecasting models is taken as the system state equation in EnKF.

Step 10: Test model. Choose three error test standards to evaluate the accuracy of models. The mean absolute percentage error (MAPE) is the most important criterion for evaluating error. Moreover, standard

deviation (STD) and root mean square percentage error (RMSPE) are also given. The detailed definitions are as follows:

(1) MAPE

$$MAPE = \frac{1}{N} \sum_{k=1}^N \left| \frac{\hat{x}^{(0)}(k) - x^{(0)}(k)}{x^{(0)}(k)} \right| \times 100\% \quad (19)$$

(2) STD

$$STD = \sqrt{\frac{1}{N} \sum_{k=1}^N \left(\left| \frac{\hat{x}^{(0)}(k) - x^{(0)}(k)}{x^{(0)}(k)} \right| - MAPE \right)^2} \quad (20)$$

(3) RMSPE

$$RMSPE = \sqrt{\frac{1}{N} \sum_{k=1}^N \left(\frac{\hat{x}^{(0)}(k) - x^{(0)}(k)}{x^{(0)}(k)} \right)^2} \times 100\% \quad (21)$$

Step 11: Forecasting the battery RUL and evaluating its accuracy.

Step 12: Comparison analysis. Choosing general and typical single prediction models as comparative methods and ablation study is conducted.

4. Case study

4.1. Dataset description

Data came from four batteries B5, B6, B7, and B18 provided by NASA Ames Prognostics Center of Excellence. After a preliminary look at the public dataset, it is found that capacity regeneration has occurred in all four batteries.

In this paper, the prediction length is set as 70, in which the prediction length of the stable normal degradation process is 50 and that of the capacity regeneration process is 20. According to the dataset provided by NASA, three batteries except B18 have 168 cycles in total. Therefore, the design of the same prediction length can show the performance of the hybrid grey-EnKF method under different data lengths.

4.2. Results analysis

In this section, the RUL of batteries B5, B6, B7, and B18 are predicted by the hybrid grey-EnKF approach. At the same time, single grey forecasting models and filtering algorithms are selected as comparative models to show the improvement effect of the hybrid grey-EnKF approach. In the framework of grey model, we choose the most classic continuous GM(1,1) and discrete non-homogeneous grey prediction model, abbreviated as GM and NDGM. For filtering algorithms, KF and EnKF are selected to show effects.

First, select the B5 data as the test object. Fig. 4 is the effects of filter algorithms. Here, to understand the advantage of the filtering algorithm, only the combination of original sequence and noise is selected as the system state equation, which does not include the state transition of any model or method. The prediction results obtained by KF and EnKF algorithms surround the entire original sequence, where KF results are disorderly and its predicted value fluctuates with the influence of noise level. Since EnKF is no longer restricted by the noise that must meet the Gaussian distribution and linear model, the prediction results of the EnKF algorithm fully conform to the degradation trend of B5, which also verifies the effectiveness and reliability of choosing EnKF as a part of the hybrid grey-EnKF.

The forecasting results for B5 are shown in Fig. 5. It can be seen that GM is in line with the degradation trend of B5 on the whole, but it cannot reflect a specific nonlinear change in battery degradation. There are some errors in cycles that reach the battery failure threshold. Compared with GM, NDGM is more flexible. The non-homogeneous type of NDGM makes the predicted curve not only an approximately straight line but also reflects some nonlinearity of battery degradation. At the

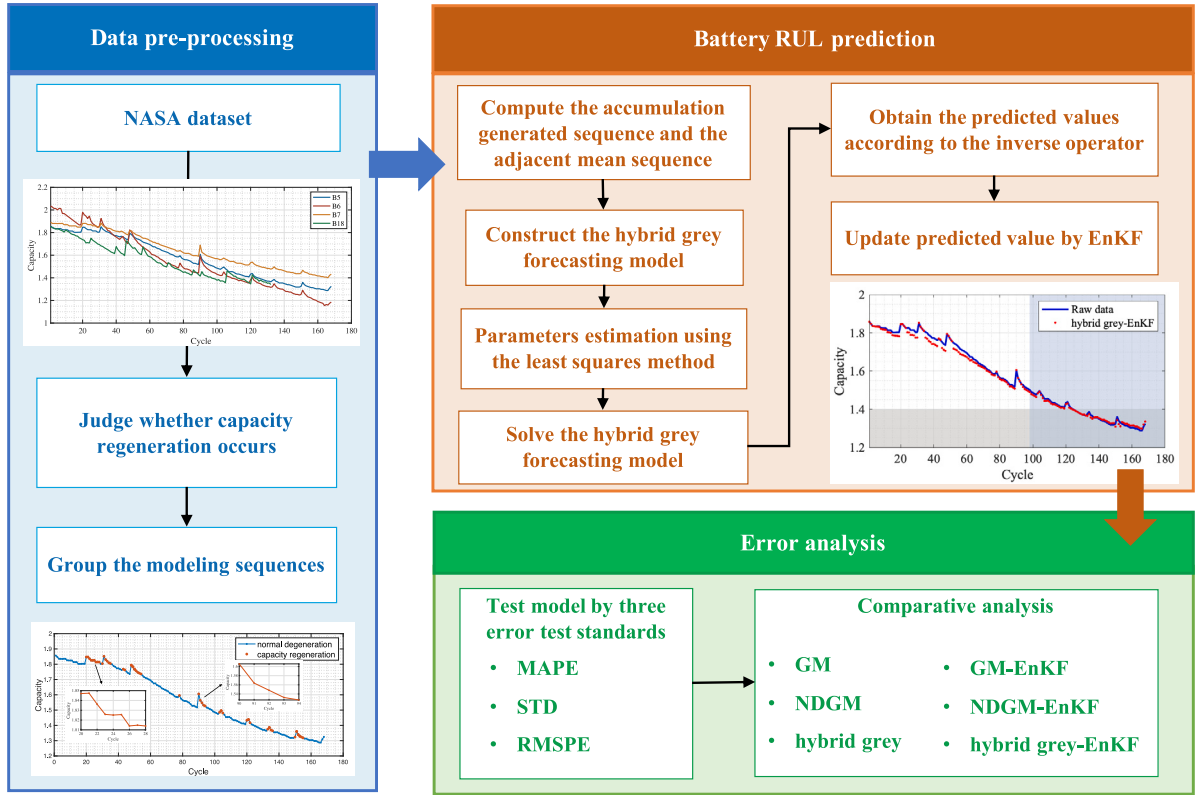


Fig. 3. Frameworks of hybrid grey-EnKF approach.

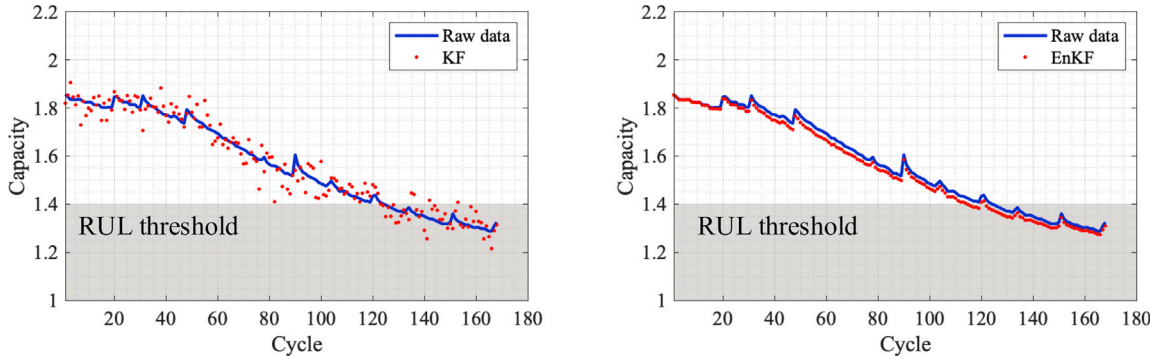


Fig. 4. Effects by filtering algorithm.

same time, the number of battery cycles that reach the failure threshold comes near the real result. However, as the prediction step increases, its reliability decreases. Compared with the single grey prediction model, the battery degradation trend of the hybrid grey model of GM and NDGM improves its accuracy and can reflect the capacity regeneration phenomenon in the whole degradation range. That is very significant in the RUL prediction of lithium batteries. However, it is disappointing that the prediction at some points is invalid.

Although the prediction performance of the hybrid grey model in Fig. 5(e) is satisfactory, for a certain length of the prediction step, the grey prediction model may have some prediction failure. As shown in Fig. 5(a),(c), with the increase of the prediction step, when the cycle is between 90 and 100, the capacity prediction value has a large error. Although the battery is in a failed state at this time, the result could be ignored, to better predict the whole cycle process, the filter

algorithm is integrated for online update. Further, Fig. 5(b)(d)(f) is the RUL prediction result of grey models with filter algorithm for B5. With the help of EnKF, by contrast, GM-EnKF and NDGM-EnKF perform some corrections for single models, which improves accuracy to some extent and cannot detailedly reflect the nonlinear characteristics of battery degradation. The hybrid grey-EnKF performs extremely well both in terms of the accuracy of the method and the trend of battery capacity degradation.

Moreover, the experiments are carried out on batteries 6,7,18. The RUL prediction results are shown in Figs. 6–8, respectively. Detailed errors are shown in Table 1. Fig. 9 is the APEs at each point for six models. To some extent, all methods listed in the comparison have satisfactory prediction results. The cycle when the battery reaches the 70% rate capacity is close to the real value. However, the single grey prediction model and filter algorithm cannot predict the trend

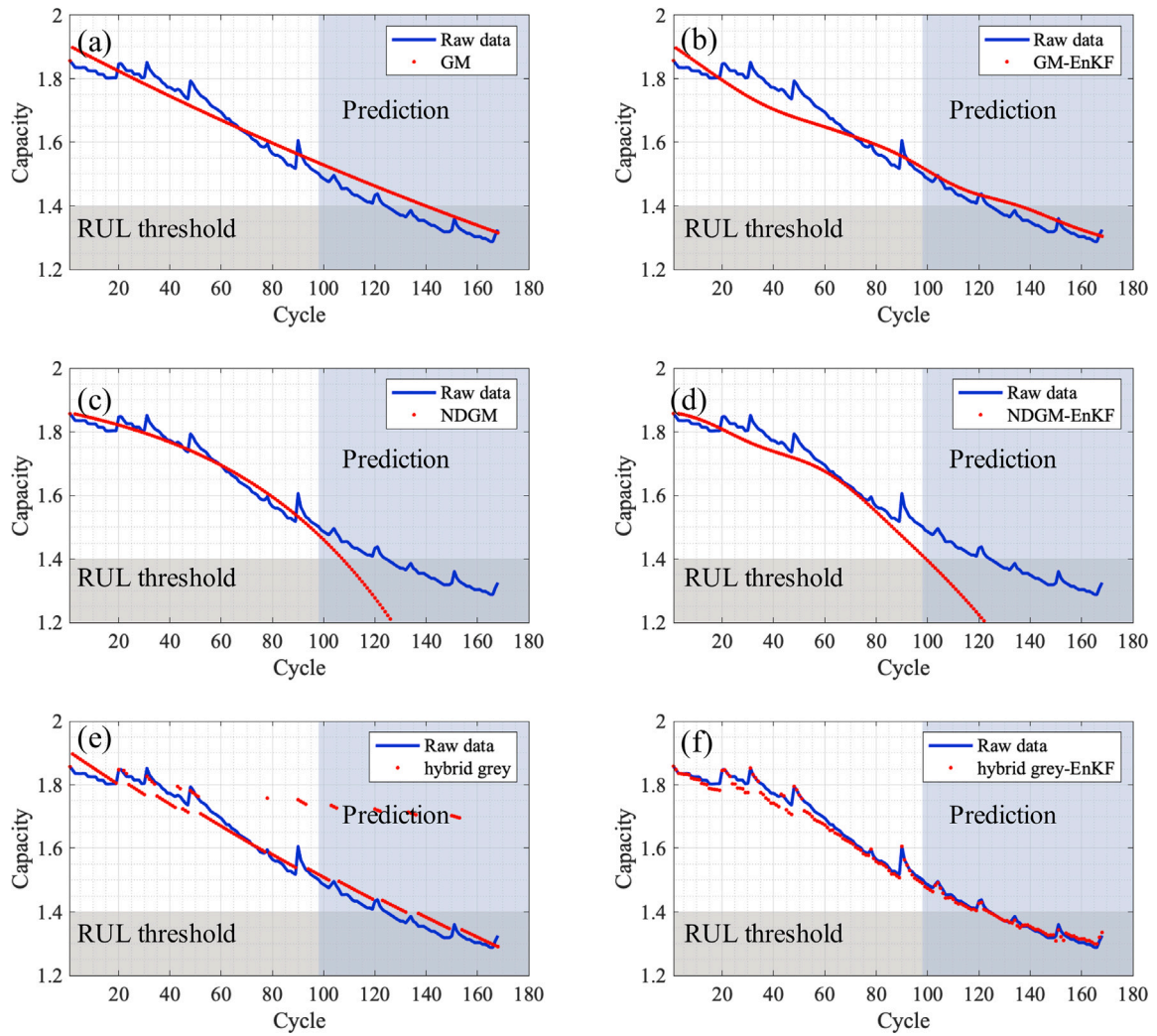


Fig. 5. Forecasting results for B5.

Table 1
Prediction accuracy evaluation.

		GM	NDGM	Hybrid-grey	KF	EnKF	GM-EnKF	NDGM-EnKF	Hybrid grey-EnKF
B5	MAPE	2.3627	10.6465	4.1686	2.2853	1.8745	1.8974	11.3060	0.6516
	STD	2.3391	10.5413	4.1280	2.2625	1.8557	1.8784	11.1941	0.6451
	RMSPE	0.0263	0.1978	0.1050	0.0295	0.0188	0.0236	0.2006	0.8266
B6	MAPE	3.2129	4.4577	6.2302	2.4527	1.8727	4.7118	5.1248	0.7954
	STD	3.1808	4.4139	6.1691	2.4282	1.8540	4.6648	5.0743	0.7875
	RMSPE	0.0389	0.0920	0.1344	0.0313	0.0189	0.0586	0.0969	1.0531
B7	MAPE	1.1449	7.0294	2.6936	1.9866	1.8858	2.7146	8.6375	0.9300
	STD	1.1334	6.9600	2.6680	1.9668	1.8669	2.6875	8.5519	0.9207
	RMSPE	0.0136	0.1337	0.0887	0.0261	0.0189	0.0300	0.1431	1.0907
B18	MAPE	2.0803	6.8903	4.2279	2.5627	1.7919	1.6875	5.9834	0.7150
	STD	2.0596	6.8221	4.1868	2.5371	1.7740	1.6706	5.9243	0.7079
	RMSPE	0.0256	0.1197	0.1071	0.0314	0.0180	0.0211	0.1105	1.0307

under the statement. The capacity regeneration phenomenon cannot be reflected in the traditional method. It can be seen that the hybrid grey-EnKF can perfectly predict the degradation process, which coincides with the true value.

Similar to the prediction effect of B5, in predictions of batteries 6, 7 and 18, the prediction of GM is similar to a straight line, and the non-homogeneity makes NDGM different. As the predicted step increases, the predicted result diverges upward or downward from the original data. Through the method in this experiment, after the separation and processing of capacity data, the effect of GM-NDGM

can roughly reflect the capacity regeneration phenomenon, but some data have large errors. Under the same noise setting, EnKF has better performance than KF.

Then, after the choice of EnKF algorithm, different from the traditional single grey model, the performance of GM-EnKF and NDGM-EnKF may improve the accuracy. Sometimes, in Fig. 6(b), EnKF algorithm does not play a positive role in the results of a single model, but makes the junction error larger. This is because the filtering algorithm itself has the noise setting, while the grey prediction model does not consider it. Combined with GM and NDGM, hybrid grey-EnKF solves

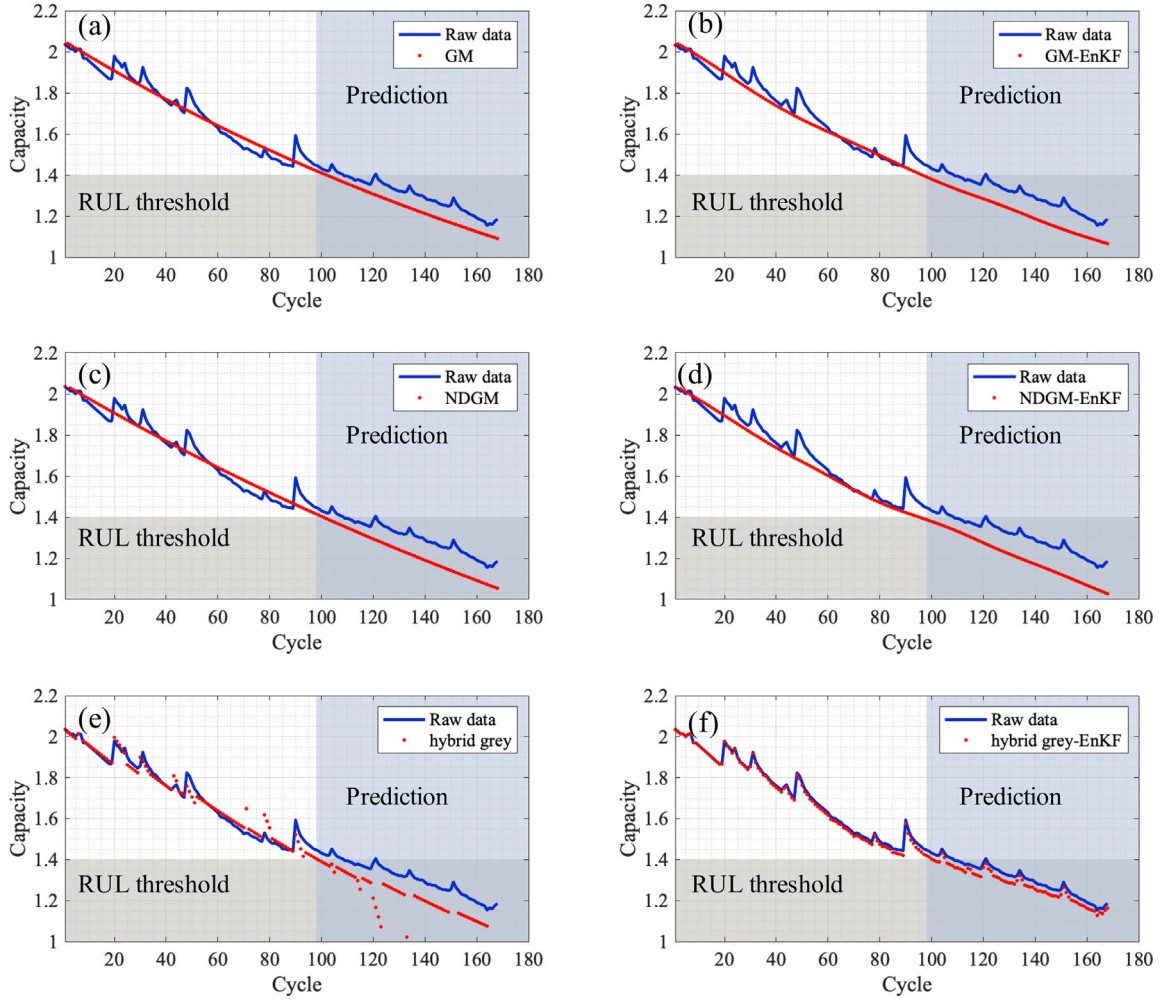


Fig. 6. Forecasting results for B6.

the problem of partial failure of GM-NDGM model in the process of prediction, which has an excellent ability to predict lithium-ion battery RUL. One thing worth noting is that due to the disturbance of random noise in EnKF, there are some subtle changes in the results of each run, and Figs. 5–8 presented is only the result of one of them.

The RUL results for batteries are listed in Table 2. Here, the cycle that is closest to the true RUL is found in the prediction. It can be seen that the prediction results based on the hybrid grey-EnKF are effective and reliable. The predictive RUL is slightly ahead of the actual value, which effectively avoids serious accidents after battery failure and makes full use of battery cycles to decrease excessive waste.

Computing speed is indeed an important indicator to evaluate algorithms. As a special data-driven method, grey prediction models are different from other big-data machine learning. Modeling based on small sample data is the greatest feature and advantage of the grey model. The prediction results of the grey model are obtained immediately. And its computing time can be completely ignored, which only takes 0.09 s. This is also an important reason why the grey model can occupy a place in the prediction algorithm with its simple modeling mechanism and fast computing speed. In addition, there is no hyperparameter and noise in the grey model, and all data in the training set are used in the model, so its robustness does not matter much.

4.3. Analysis of ablation experiment

The method proposed in this paper is a hybrid algorithm composed of GM, NDGM and EnKF. The ablation study further clarifies the

specific role of different modules in the method. In order to reflect the degree of contribution of the three components to the method, a one-to-one comparison between the two is carried out. Due to the disturbance effect of random noise in EnKF, we conducted 100 repeated experiments. Take MAPE as the criterion for judgment, and take the average value of 100 runs as the general accuracy of each model.

Fig. 10 shows the comparison of hybrid grey-EnKF and its degenerate models. It can be found that compared with its generated model, the performance of the proposed hybrid grey-EnKF has been greatly improved. For the NDGM series, the accuracy is improved by 87.52% and 87.70%. This is because throughout the battery degradation stage, the capacity regeneration only occurs in a few charging and discharging cycles, and most of the time the battery is in a stable degradation stage. For GM, adding EnKF is helpful for accuracy, but it is only slightly improved. Compared with hybrid grey-EnKF, the accuracy improvement of GM-EnKF is the smallest, which is 27.09%. It can be seen that GM-EnKF occupies a large place in the proposed model.

After clarifying the importance of the model module, the input of the model is also another key point affecting the accuracy. In this case study, the model input is only the battery capacity sequence. Therefore, we change the length of the input sequence, that is, set different prediction step lengths, to verify the stability of the method proposed. Fig. 11 shows the comparison of hybrid grey-EnKF and GM-EnKF under different prediction steps. It can also be verified from Fig. 11 that the overall performance of hybrid grey-EnKF is higher than that of GM-EnKF. Under the setting of different prediction steps, the model also has excellent performance. When the predicted step length

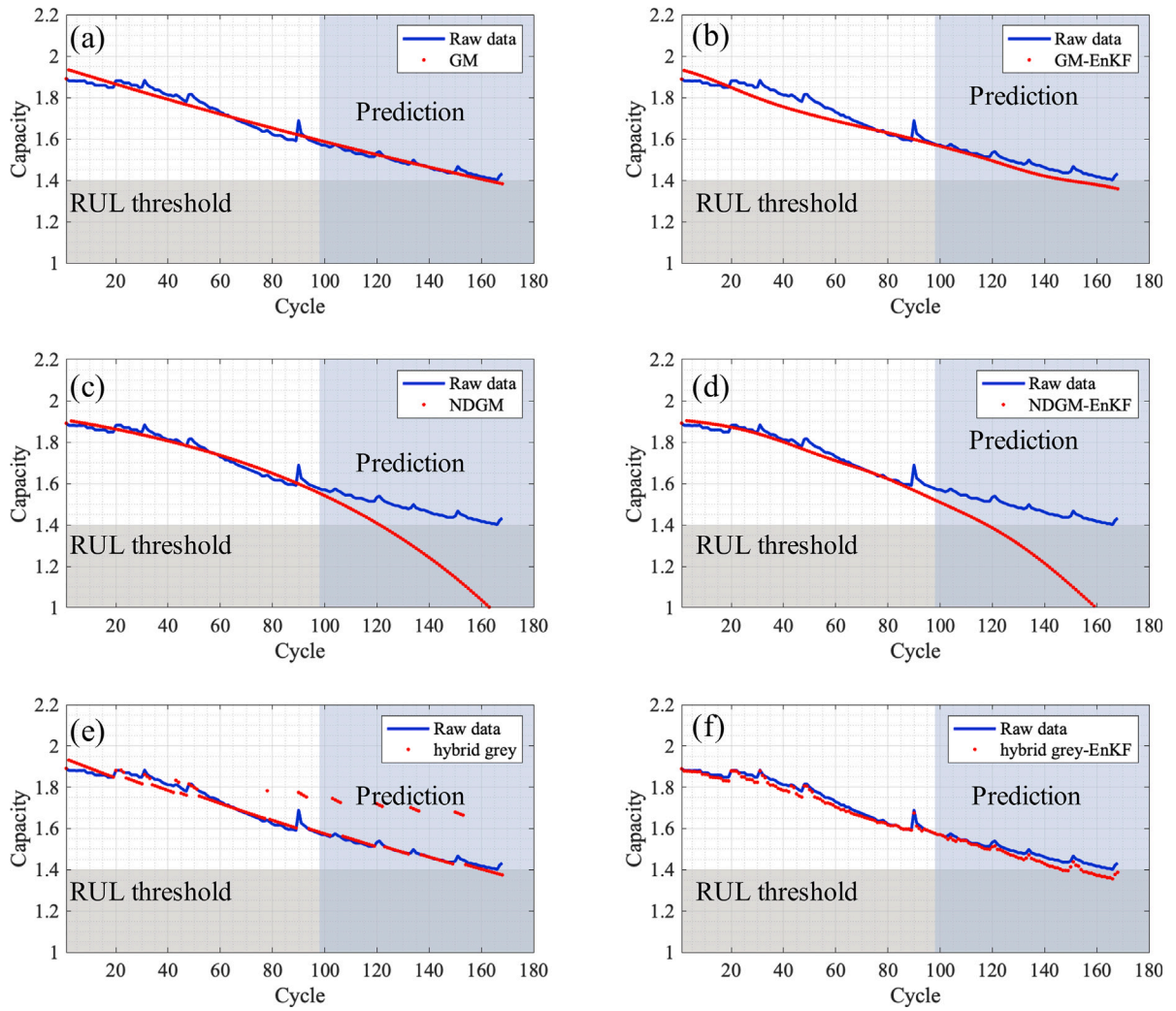


Fig. 7. Forecasting results for B7.

Table 2
RUL results for batteries.

		True	GM	NDGM	Hybrid-grey	KF	EnKF	GM-EnKF	NDGM-EnKF	Hybrid grey-EnKF
B5	EOL	124	139	107	91	99	114	122	106	123
	Capacity	1.4012	1.4024	1.4024	1.4644	1.4158	1.4006	1.4004	1.4062	1.4045
B6	EOL	108	101	100	98	94	101	95	98	107
	Capacity	1.4049	1.4047	1.4040	1.4044	1.4062	1.4024	1.4068	1.4033	1.4001
B7	EOL	166	161	121	160	128	156	146	117	156
	Capacity	1.4005	1.4027	1.4005	1.4031	1.4060	1.4018	1.4017	1.4026	1.4054
B18	EOL	95	113	–	100	102	89	107	–	94
	Capacity	1.4057	1.4006	–	1.4053	1.4065	1.4003	1.4021	–	1.4077

increases, its accuracy does not decrease significantly. This also greatly verifies the effectiveness and correctness of this paper combined with the EnKF algorithm to eliminate the error of the grey prediction model in long-term prediction.

4.4. Comparative analysis with other methods

The proposed hybrid grey-EnKF method is a kind of data-driven prediction method. Further, the RUL prediction results of the existing data-driven methods for battery capacity are compared. Table 3 lists the predictions from existing studies. The methods listed in Table 3 include common machine learning methods, as well as hybrid methods

of machine learning models and intelligent optimization algorithms, so as to compare the methods proposed in this paper in a more comprehensive way.

As we all know, the ability of machine learning depends on abundant high-quality data and parameter adjustment. The search optimization of the parameters and the determination of the hyper-parameters have a great influence on its accuracy. Hence, relevant parameters should be set first. For common machine learning methods listed in Table 3, such as BLSTM, LSTM, GRU, RNN, we keep the default parameters under the package framework as much as possible. For hybrid methods of machine learning models and intelligent optimization algorithms, considering that methods themselves have been improved,

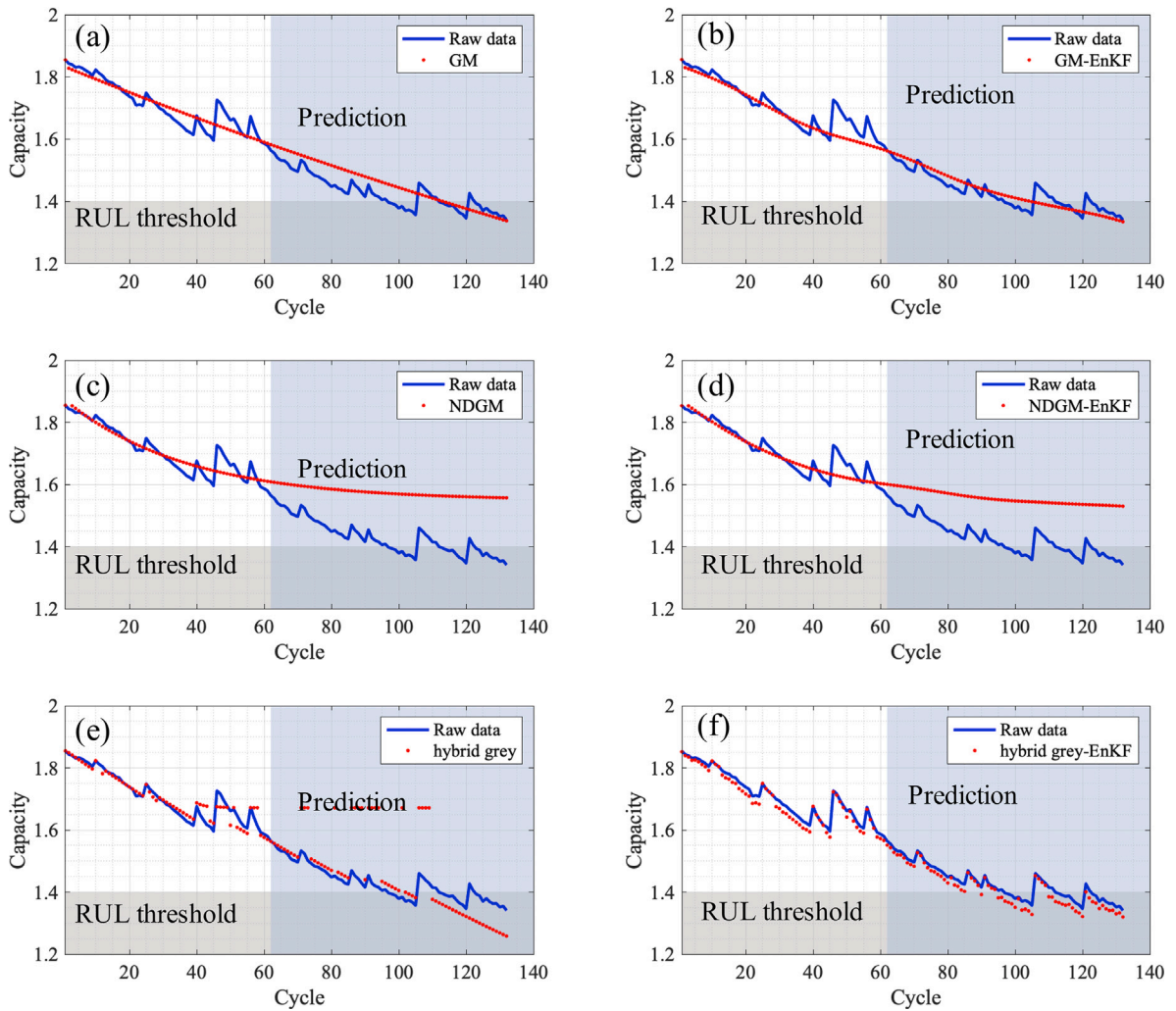


Fig. 8. Forecasting results for B18.

Table 3
Comparison with existing data-driven methods.

Method	B5 MAPE (%)	B6 MAPE (%)	B7 MAPE (%)	B18 MAPE (%)
CLGPFR	1.60	10.20	1.70	—
SE-MGPR	1.38	2.93	1.02	—
RTPF	0.76	1.25	0.44	—
PSO-SVR	0.82	1.25	1.02	—
VMD-GPR	1.30	—	—	—
VMD-PF	1.89	—	—	—
PSR-SVR	1.54	0.06	1.39	1.75
PSR-GASVR	1.30	0.82	0.95	1.16
PSR-GASVR-EC	0.21	0.09	1.05	0.27
BLSTM	5.97	10.89	—	5.19
LSTM	9.57	12.52	—	3.87
GRU	11.17	8.23	—	6.53
RNN	7.11	10.49	—	12.33
Hybrid grey-EnKF	0.65	0.80	0.93	0.72

we have drawn on the methods and their predictive performance from the corresponding references on the same dataset. Excessive parameter optimization is not a key focus in this paper. However, the proposed method in this paper does not have a parameter adjustment process. Except for the setting of random noise, no parameters need to be adjusted during the whole training and prediction stages. This also reflects the great advantages of simple modeling and the fast running speed of the proposed method.

As can be seen from Table 3, the performance of the grey-EnKF in this paper is better than half of the other methods, which is almost the same as that of RTPF (He et al., 2015) and PSO-SVR (Qin et al., 2015). Only the precision of the PSR-GASVR-EC (Chen et al., 2020) method is far higher than that of the hybrid grey-EnKF method, but its complexity is also far higher than that of the grey-EnKF. Moreover, signal processing is also a common method in the current literature considering the capacity regeneration phenomenon (Zhang et al., 2022). Signal decomposition such as EMD and CEEMDAN decompose the capacity sequence into several sub-signal sequences, and then reassemble the signals after prediction (Chen et al., 2021a; Liu et al., 2021; Meng, 2022). Such methods lack some realistic interpretation while possessing excellent prediction accuracy. Although this paper also divides capacity into normal degradation and regeneration sequences, this kind of decomposition is more in line with the practical operating logic in the real situation.

5. Conclusion

This work focuses on the detailed degradation process in lithium-ion batteries so that a novel hybrid grey-EnKF approach is proposed. The main conclusions are as follows:

(1) This paper proposed a hybrid grey-EnKF approach that can develop a novel idea to address RUL prediction of lithium-ion batteries with capacity regeneration, which combines non-iterative structure models and iterative filtering algorithms.

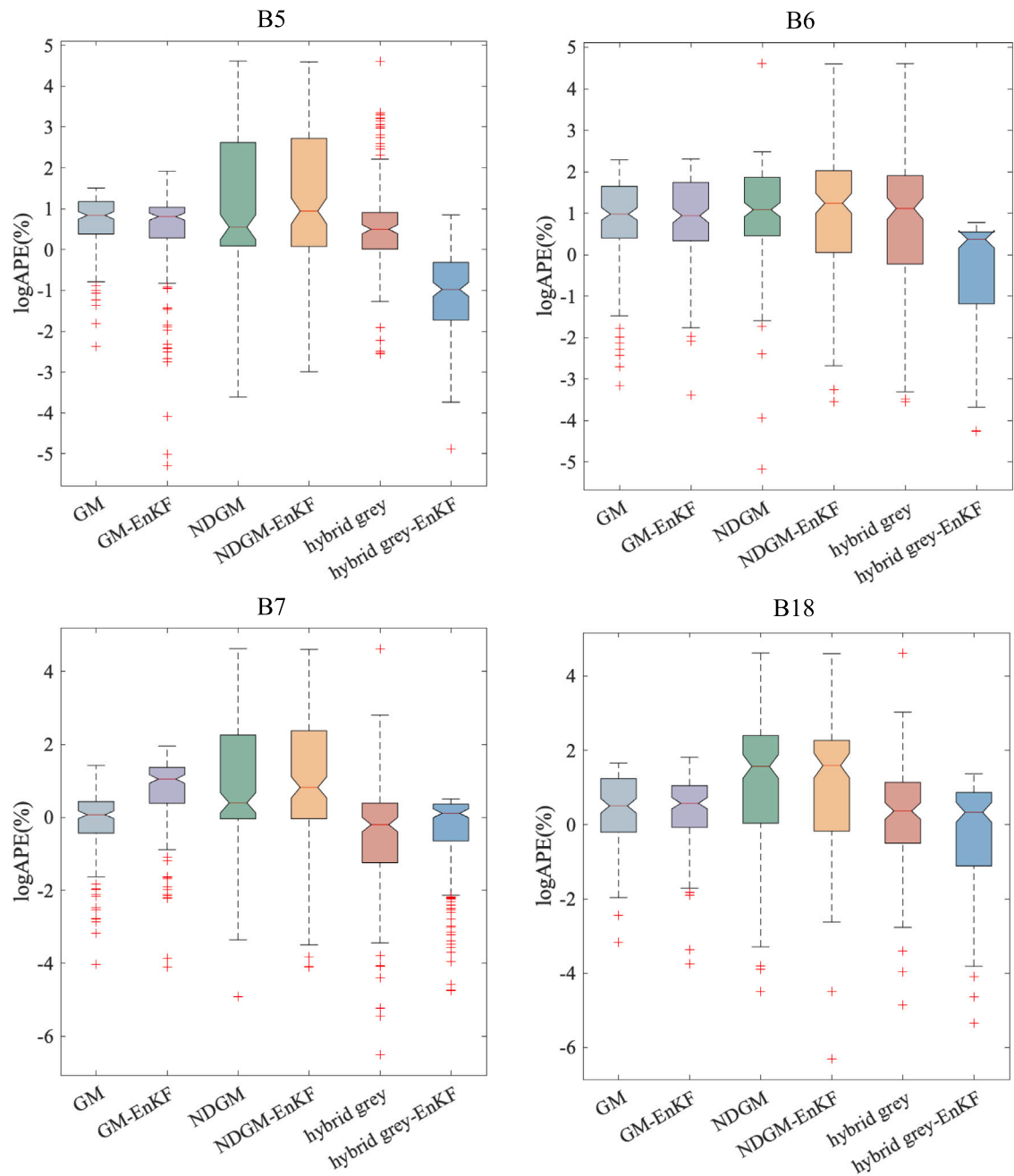


Fig. 9. APEs for different models.

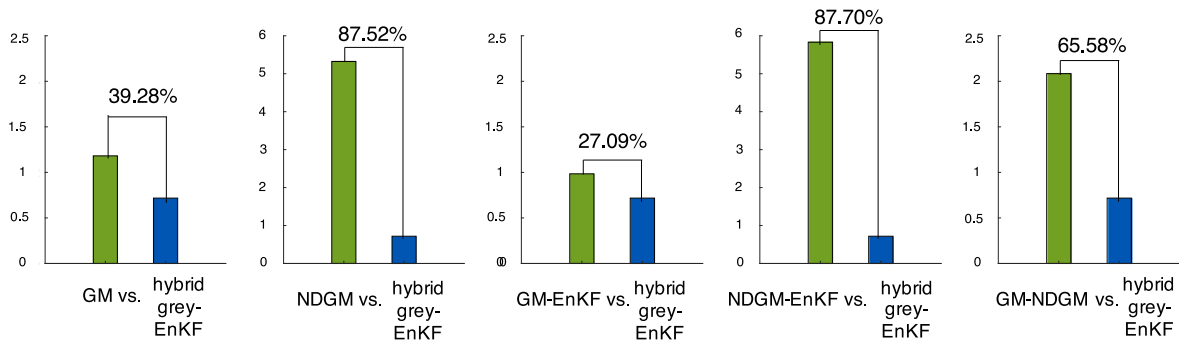


Fig. 10. Comparison of hybrid grey-EnKF and its degenerated models regarding MAPE.

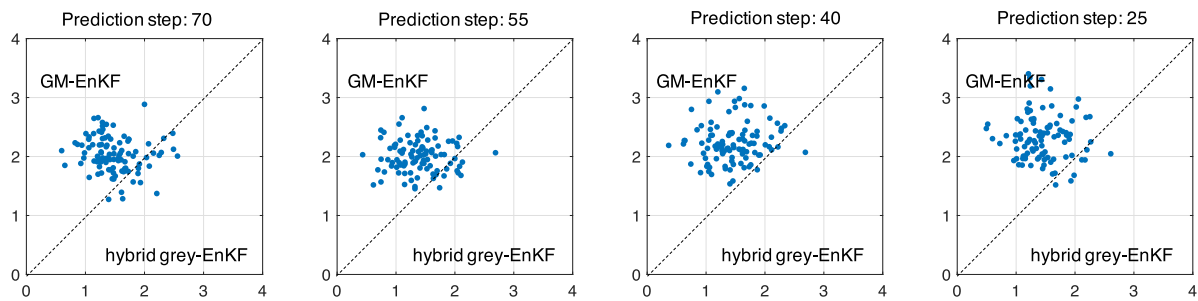


Fig. 11. Comparison of hybrid grey-EnKF and GM-EnKF under different prediction steps.

(2) Without the framework of machine learning and signal decomposition, the hybrid grey forecasting model is modeled according to the degradation characteristics of the normal degradation and regeneration phenomena at different stages, which does not need abundant data and has little cost to take the real degradation process into account. EnKF helps to neutralize the side effects of grey forecasting models for long-term prediction.

(3) For battery data with different lengths, the proposed hybrid grey-EnKF approach demonstrates a considerably satisfactory prediction performance of RUL, whose accuracy is higher than most hybrid machine learning models.

Some considerations about the proposed approach are worth further putting into account. In EnKF framework, the state equation depending on the core prediction model is a key issue. Grey models are non-iterative structural forecasting models, so we need to ensure the accuracy of grey models as a precondition. The grey model in this paper can be further expanded to a generalized nonlinear and non-homogeneous structure to adapt to more complex battery degradation laws. Moreover, with the ability of grey modeling to mine sequence information, it can greatly show the greatest advantage of the two in combination with the framework of machine learning.

CRedit authorship contribution statement

Kailing Li: Conceptualization, Methodology, Programming, Writing – original draft. **Naiming Xie:** Funding acquisition, Supervision, Writing – review & editing. **Hui Li:** Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by National Natural Science Foundation of China under grant 72171116, 72301140 and 92367301, the Fundamental Research Funds for the Central Universities, China under grant NK2023001, “333 talent” project in Jiangsu Province (China) and Postgraduate Research & Practice Innovation Program of Jiangsu Province under grant KYCX24_0513, KYCX24_0520 (China).

Data availability

The data in this paper is based on the publicly available datasets published by NASA Prognostics Center of Excellence.

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